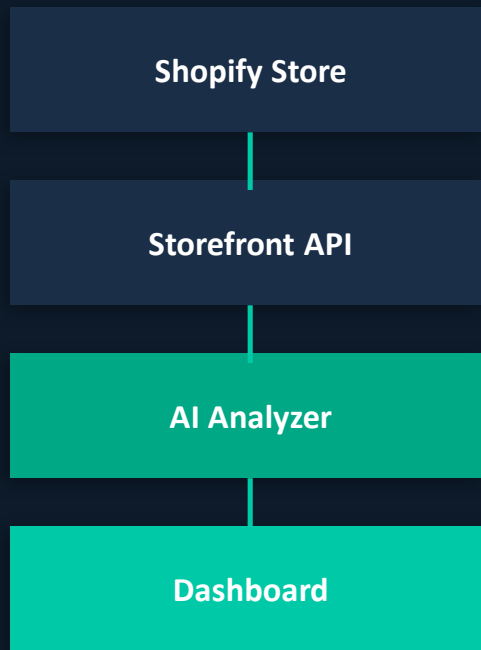


Shopify AI Readiness Analyzer

An AI-powered platform that evaluates Shopify stores for trust, clarity, and AI-driven commerce readiness.



AI Powered

Shopify API

LLM Analysis

Real-time Scores

Problem & Solution

⚠️ PROBLEM

- ✗ Unclear product descriptions
- ✗ Missing store policies
- ✗ Poor trust indicators
- ✗ Incomplete FAQs
- ✗ Non-AI-friendly content

↓ Customer Trust

↓ Conversions

↓ AI Discovery



✓ SOLUTION

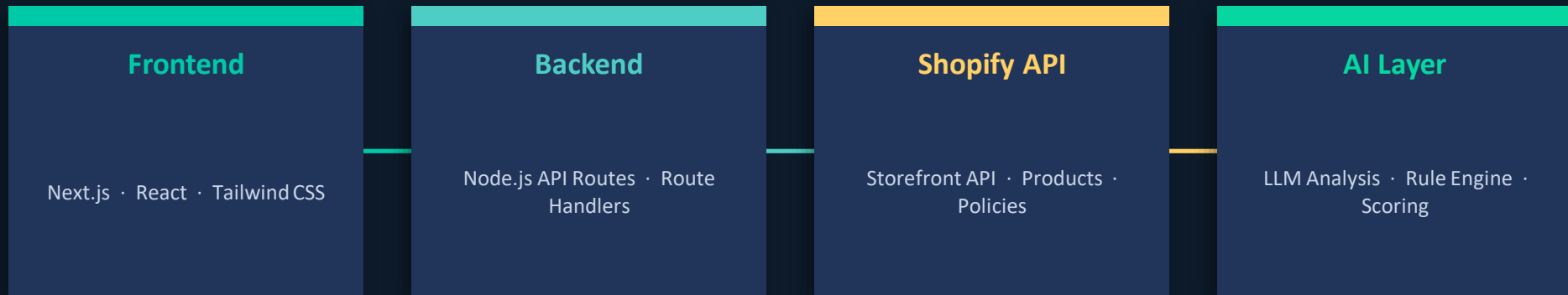
1. Connect to Shopify stores
2. Fetch store data via API
3. Analyze content quality
4. Detect missing information
5. Generate AI recommendations

↑ Trust Score

↑ Conversions

↑ AI-Ready

System Architecture



DATA FLOW



Next.js

React

Tailwind

Node.js

Shopify API

Claude AI

TypeScript

Shopify API Integration

What We Fetch

Product Titles

Descriptions

Product Images

Store Policies

FAQs / Pages

Why Shopify Storefront API?

⚡ Structured, clean ecommerce data

🔗 Fast, reliable integration

📈 Scalable architecture

🚫 No manual scraping needed

Key Decision: Shopify Storefront API provides clean, reliable, and structured ecommerce data directly from Shopify stores — enabling consistent AI analysis.

AI Analyzer & Core Logic

⚙️ Deterministic Logic

- ▶ Missing field detection
- ▶ Empty description checks
- ▶ Duplicate content flags
- ▶ Policy presence check
- ▶ Keyword validation

Fast · Consistent · Deterministic

VS

🧠 AI / LLM Analysis

- ▶ Readability & clarity scoring
- ▶ Trust signal evaluation
- ▶ Ambiguity detection
- ▶ Recommendation generation
- ▶ AI-readiness scoring

Contextual · Intelligent · Adaptive

Hybrid Architecture = Reliability + Consistency + Contextual Intelligence

Failure Handling & System Reliability

🔥 Shopify API Failure

- ✓ Retry mechanism (up to 3x)
- ✓ Fallback error handling
- ✓ Graceful degradation to cached data

⚠️ Invalid User Input

- ✓ URL format validation
- ✓ Input sanitization layer
- ✓ Structured backend error responses

📄 Malformed AI Response

- ✓ Retry AI call once
- ✓ Fallback deterministic scoring
- ✓ Cached recommendation templates

🛡️ System Degradation

- ✓ Partial results shown to user
- ✓ System remains functional
- ✓ Clear status feedback provided

System remains functional even when external services partially fail.

Security & Technical Decisions

Security Measures

 Server-side API calls only

 Protected environment variables

 Sanitized & validated inputs

 Controlled API access policies

Key Technical Decisions

Next.js Full-stack rapid development

Tailwind CSS Scalable and consistent UI

Hybrid AI Stability + contextual intelligence

API Architecture Modularity and scalability

AI ↔ Deterministic Boundary

AI handles contextual understanding, nuance & recommendations · Deterministic logic handles fixed validations, field checks & policy detection

Limitations & Future Improvements

Current Limitations

- ✗ Shopify-only support
- ✗ Limited multilingual capability
- ✗ AI output variability
- ✗ Dependent on available store data

Future Improvements

- AI agent integration
- Automated content rewriting
- SEO optimization module
- Multilingual analysis
- Analytics dashboard
- Downloadable PDF reports
- Competitor benchmarking

Roadmap: Phase 1 → Multi-platform support | Phase 2 → Auto-rewrite engine | Phase 3 → Full analytics suite

Thank You

The Shopify AI Readiness Analyzer

Shopify API
Integration

Deterministic
Validation

AI-Powered
Analysis

Improving trust · clarity · AI discoverability · shopping experience

AI-driven ecommerce optimization at scale.